

CORE 404 DSA ROADMAP

Complete Practice Problem Master List · Grouped by Algorithmic Pattern

381

TOTAL PROBLEMS

82

PATTERNS

28

MAJOR TOPICS

108 / 224 / 49

EASY / MED / HARD

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1. Why Pattern-Based Learning?

Isolated problem memorization fails in interviews. Grouping problems by pattern builds intuitive muscle memory to identify underlying algorithm archetypes rapidly in unseen questions.

2. Roadmap Difficulty Split

108 Easy: Solidify core syntax, pointer movement & edge cases.
224 Medium: Standard Tier-1 interview bar (optimizations).
49 Hard: Multi-state DP, Segment Trees & advanced graphs.

3. Interactive Features & Tools

Every problem in this PDF has a direct clickable practice link to LeetCode / GFG. Use the accompanying **Core404_Problems_Grouped_By_Pattern.xlsx** for interactive status tracking.

Pattern 1: Foundations / Programming Basics					14 Problems · Topic: Foundations
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-001	Read and Print a Number	Easy	GeeksforGeeks	Solve on GFG
2	LR-002	Basic Data Types	Easy	GeeksforGeeks	Solve on GFG
3	LR-003	Basic Arithmetic Operations	Easy	GeeksforGeeks	Solve on GFG
4	LR-004	Check Even or Odd	Easy	GeeksforGeeks	Solve on GFG
5	LR-005	Find Largest of Three Numbers	Easy	GeeksforGeeks	Solve on GFG
6	LR-006	Print Numbers from 1 to N	Easy	GeeksforGeeks	Solve on GFG
7	LR-007	Sum of First N Natural Numbers	Easy	GeeksforGeeks	Solve on GFG
8	LR-008	Pattern Printing	Easy	GeeksforGeeks	Solve on GFG
9	LR-009	Functions in Programming	Easy	GeeksforGeeks	Solve on GFG
10	LR-010	Array Traversal	Easy	GeeksforGeeks	Solve on GFG
11	LR-011	String Traversal	Easy	GeeksforGeeks	Solve on GFG
12	LR-012	Count Digits in a Number	Easy	GeeksforGeeks	Solve on GFG
13	LR-013	Sum of Digits	Easy	GeeksforGeeks	Solve on GFG
14	LR-014	Time and Space Complexity Basics	Easy	GeeksforGeeks	Solve on GFG

Pattern 2: Math for DSA					11 Problems · Topic: Math / Number Theory
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-015	Prime Factors	Easy	GeeksforGeeks	Solve on GFG
2	LR-016	Closest Number	Easy	GeeksforGeeks	Solve on GFG
3	LR-017	Euler's Totient	Easy	GeeksforGeeks	Solve on GFG
4	LR-018	Kth Set Bit	Easy	GeeksforGeeks	Solve on GFG
5	LR-019	Only set bit	Easy	GeeksforGeeks	Solve on GFG
6	LR-020	nCr	Easy	GeeksforGeeks	Solve on GFG
7	LR-021	Check if a Number is Prime	Easy	GeeksforGeeks	Solve on GFG
8	LR-022	GCD of Two Numbers	Easy	GeeksforGeeks	Solve on GFG
9	LR-023	LCM of Two Numbers	Easy	GeeksforGeeks	Solve on GFG
10	LR-024	Count Set	Medium	GeeksforGeeks	Solve on GFG
11	LR-025	Count Primes — Sieve of Eratosthenes	Medium	LeetCode	Solve on LeetCode

Pattern 3: Fast Exponentiation					2 Problems · Topic: Math / Number Theory
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-026	Pow(x, n)	Medium	LeetCode	Solve on LeetCode
2	LR-027	Modular Exponentiation	Medium	GeeksforGeeks	Solve on GFG

Pattern 4: Arrays					24 Problems · Topic: Arrays
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-028	Linear Search	Easy	GeeksforGeeks	Solve on GFG
2	LR-029	Find Minimum and Maximum	Easy	GeeksforGeeks	Solve on GFG
3	LR-030	Third Largest Element	Easy	GeeksforGeeks	Solve on GFG
4	LR-031	Rotate Array	Easy	LeetCode	Solve on LeetCode
5	LR-032	Reverse Array in Groups	Easy	GeeksforGeeks	Solve on GFG
6	LR-033	Move All Zeroes to End	Easy	GeeksforGeeks	Solve on GFG
7	LR-034	Maximum Consecutive Ones	Easy	GeeksforGeeks	Solve on GFG
8	LR-035	Array Leaders	Easy	GeeksforGeeks	Solve on GFG
9	LR-036	Wave Array	Easy	GeeksforGeeks	Solve on GFG

#	ID	Problem Name	Difficulty	Platform	Practice Link
10	LR-037	Plus One	Easy	LeetCode	Solve on LeetCode
11	LR-038	Stock Buy and Sell — One Transaction	Easy	LeetCode	Solve on LeetCode
12	LR-039	Rearrange Array Alternately	Easy	GeeksforGeeks	Solve on GFG
13	LR-040	Find Common Elements in Three Sorted Arrays	Easy	GeeksforGeeks	Solve on GFG
14	LR-041	Maximum Difference Between Two Elements	Easy	GeeksforGeeks	Solve on GFG
15	LR-042	Majority Element	Easy	LeetCode	Solve on LeetCode
16	LR-048	Second Largest Element	Medium	GeeksforGeeks	Solve on GFG
17	LR-049	Reverse an Array	Medium	GeeksforGeeks	Solve on GFG
18	LR-050	Sum of All Subarrays	Medium	GeeksforGeeks	Solve on GFG
19	LR-051	Maximum Subarray	Medium	LeetCode	Solve on LeetCode
20	LR-052	Majority Element II	Medium	LeetCode	Solve on LeetCode
21	LR-053	Next Permutation	Medium	LeetCode	Solve on LeetCode
22	LR-054	Maximum Circular Subarray Sum	Medium	LeetCode	Solve on LeetCode
23	LR-059	Trapping Rain Water	Hard	LeetCode	Solve on LeetCode
24	LR-060	Smallest Missing Positive Number	Hard	GeeksforGeeks	Solve on GFG

Pattern 5: Cyclic Sort / Index Placement

9 Problems · Topic: Arrays

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-043	Find all Duplicate Numbers	Easy	LeetCode	Solve on LeetCode
2	LR-044	Set Mismatch	Easy	LeetCode	Solve on LeetCode
3	LR-045	Remove Duplicates from Sorted Array	Easy	LeetCode	Solve on LeetCode
4	LR-046	Missing and Repeating in Array	Easy	GeeksforGeeks	Solve on GFG
5	LR-047	Find the Missing Number	Easy	GeeksforGeeks	Solve on GFG
6	LR-056	Cycle Sort	Medium	GeeksforGeeks	Solve on GFG
7	LR-057	Min Swaps to Sort	Medium	GeeksforGeeks	Solve on GFG
8	LR-058	First Missing Positive	Medium	LeetCode	Solve on LeetCode
9	LR-061	Find the First K Missing Positive Numbers	Hard	LeetCode	Solve on LeetCode

Pattern 6: Two Pointers

6 Problems · Topic: Arrays

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-055	Union of 2 Sorted	Medium	GeeksforGeeks	Solve on GFG
2	LR-095	Two Sum II - Sorted Array	Medium	LeetCode	Solve on LeetCode
3	LR-371	Valid Palindrome II	Easy	LeetCode	Solve on LeetCode
4	LR-096	Sort Colors	Medium	LeetCode	Solve on LeetCode
5	LR-097	3Sum	Medium	LeetCode	Solve on LeetCode
6	LR-098	Container With Most Water	Medium	LeetCode	Solve on LeetCode

Pattern 7: Strings

4 Problems · Topic: Strings

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-062	Manacher's Algorithm	Easy	GeeksforGeeks	Solve on GFG
2	LR-064	Same Strings	Medium	GeeksforGeeks	Solve on GFG
3	LR-065	Multiply Large Numbers	Medium	GeeksforGeeks	Solve on GFG
4	LR-066	URLify	Medium	GeeksforGeeks	Solve on GFG

Pattern 8: String Matching

4 Problems · Topic: Strings

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-063	Longest Prefix Suffix	Easy	GeeksforGeeks	Solve on GFG
2	LR-070	Min Repeats for Substring	Hard	GeeksforGeeks	Solve on GFG
3	LR-363	Repeated Substring Pattern	Easy	LeetCode	Solve on LeetCode
4	LR-364	Repeated String Match	Medium	LeetCode	Solve on LeetCode

Pattern 9: String Manipulation

6 Problems · Topic: Strings

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-358	Reverse String	Easy	LeetCode	Solve on LeetCode
2	LR-359	Reverse Words in a String III	Easy	LeetCode	Solve on LeetCode
3	LR-360	Rotate String	Easy	LeetCode	Solve on LeetCode
4	LR-361	String Compression	Medium	LeetCode	Solve on LeetCode
5	LR-366	Zigzag Conversion	Medium	LeetCode	Solve on LeetCode
6	LR-368	Text Justification	Hard	LeetCode	Solve on LeetCode

Pattern 10: Palindrome Patterns

4 Problems · Topic: Strings

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-067	Longest Palindromic Substrings	Medium	LeetCode	Solve on LeetCode
2	LR-068	Palindromic Substrings	Medium	LeetCode	Solve on LeetCode
3	LR-069	Palindrome	Medium	LeetCode	Solve on LeetCode
4	LR-071	Palindrome Substring Queries	Hard	GeeksforGeeks	Solve on GFG

Pattern 11: KMP Pattern Matching

1 Problem · Topic: Strings

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-362	Find the Index of the First Occurrence in a String	Easy	LeetCode	Solve on LeetCode

Pattern 12: String Manipulation / Parsing

2 Problems · Topic: Strings

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-365	String to Integer (atoi)	Medium	LeetCode	Solve on LeetCode
2	LR-369	Valid Number	Hard	LeetCode	Solve on LeetCode

Pattern 13: String Design / Serialization

1 Problem · Topic: Strings

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-367	Encode and Decode Strings	Medium	LeetCode	Solve on LeetCode

Pattern 14: Hashing

12 Problems · Topic: Hashing

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-072	Roman to Integer	Easy	LeetCode	Solve on LeetCode
2	LR-073	Separate Chaining	Easy	GeeksforGeeks	Solve on GFG
3	LR-074	Anagrams	Easy	LeetCode	Solve on LeetCode
4	LR-075	Most Frequent	Easy	GeeksforGeeks	Solve on GFG
5	LR-076	Two Sum	Easy	LeetCode	Solve on LeetCode
6	LR-077	Contains Duplicate	Easy	LeetCode	Solve on LeetCode
7	LR-078	Isomorphic Strings	Easy	LeetCode	Solve on LeetCode
8	LR-079	Group Anagrams	Medium	LeetCode	Solve on LeetCode
9	LR-080	Subarray Sum Equals K	Medium	LeetCode	Solve on LeetCode

#	ID	Problem Name	Difficulty	Platform	Practice Link
10	LR-081	Longest Consecutive Sequence	Medium	LeetCode	Solve on LeetCode
11	LR-082	4Sum II	Medium	LeetCode	Solve on LeetCode
12	LR-083	Longest Substring with At Most Two Distinct Characters	Medium	LeetCode	Solve on LeetCode

Pattern 15: Frequency Counting

1 Problem · Topic: Hashing

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-370	First Unique Character in a String	Easy	LeetCode	Solve on LeetCode

Pattern 16: Sorting

11 Problems · Topic: Sorting

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-084	Sort by Length	Easy	GeeksforGeeks	Solve on GFG
2	LR-085	Wave Form	Easy	GeeksforGeeks	Solve on GFG
3	LR-086	Is Sorted	Easy	GeeksforGeeks	Solve on GFG
4	LR-087	Kth of two sorted	Easy	GeeksforGeeks	Solve on GFG
5	LR-088	Median of two sorted	Easy	GeeksforGeeks	Solve on GFG
6	LR-089	Merge Overlapping	Medium	GeeksforGeeks	Solve on GFG
7	LR-090	Shop in Candy Store	Medium	GeeksforGeeks	Solve on GFG
8	LR-091	Intersection of 2 Sorted	Medium	GeeksforGeeks	Solve on GFG
9	LR-092	Merge Without Extra Space	Medium	GeeksforGeeks	Solve on GFG
10	LR-093	Quick Sort	Medium	GeeksforGeeks	Solve on GFG
11	LR-094	Sort Binary	Medium	GeeksforGeeks	Solve on GFG

Pattern 17: Sliding Window

18 Problems · Topic: Sliding Window

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-099	Subarrays Less Than K	Medium	GeeksforGeeks	Solve on GFG
2	LR-100	Subarrays with k odds	Medium	GeeksforGeeks	Solve on GFG
3	LR-101	First negative in every window	Medium	GeeksforGeeks	Solve on GFG
4	LR-102	Min Swaps to group all 1's	Medium	GeeksforGeeks	Solve on GFG
5	LR-103	Number of Subarrays having Average Greater or Equal to Threshold	Medium	LeetCode	Solve on LeetCode
6	LR-104	Repeated DNA sequences	Medium	LeetCode	Solve on LeetCode
7	LR-105	Longest Substring Without Repeating Characters	Medium	LeetCode	Solve on LeetCode
8	LR-106	Minimum Window Subsequence	Medium	GeeksforGeeks	Solve on GFG
9	LR-107	Smallest containing 0, 1 and 2	Medium	GeeksforGeeks	Solve on GFG
10	LR-108	Count Anagrams	Medium	GeeksforGeeks	Solve on GFG
11	LR-109	Full Distinct Subarrays	Medium	GeeksforGeeks	Solve on GFG
12	LR-110	Max Consecutive Ones	Medium	LeetCode	Solve on LeetCode
13	LR-111	Maximize Ones	Medium	GeeksforGeeks	Solve on GFG
14	LR-112	Subarray with given sum	Medium	GeeksforGeeks	Solve on GFG
15	LR-113	Subarray Product Less Than K	Medium	LeetCode	Solve on LeetCode
16	LR-114	Max Subarray Length with K Increments	Hard	GeeksforGeeks	Solve on GFG
17	LR-115	Sliding Window Maximum	Hard	GeeksforGeeks	Solve on GFG
18	LR-116	Minimum Window Substring	Hard	LeetCode	Solve on LeetCode

Pattern 18: String + Sliding Window

1 Problem · Topic: Sliding Window

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-372	Find All Anagrams in a String	Medium	LeetCode	Solve on LeetCode

Pattern 19: Prefix Sum / Prefix XOR				8 Problems · Topic: Prefix / Difference	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-117	Count with given XOR	Easy	GeeksforGeeks	Solve on GFG
2	LR-118	Compute before Matrix	Medium	GeeksforGeeks	Solve on GFG
3	LR-119	Point Counts from Intervals	Medium	GeeksforGeeks	Solve on GFG
4	LR-120	Longest Span	Medium	GeeksforGeeks	Solve on GFG
5	LR-121	Original Array	Medium	GeeksforGeeks	Solve on GFG
6	LR-122	Largest with sum divisible by k	Medium	GeeksforGeeks	Solve on GFG
7	LR-123	Longest with Sum K	Medium	GeeksforGeeks	Solve on GFG
8	LR-124	Product of Array Except Self	Medium	LeetCode	Solve on LeetCode

Pattern 20: Binary Search				3 Problems · Topic: Binary Search	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-125	First Bad Version	Easy	LeetCode	Solve on LeetCode
2	LR-126	Sqrt(x)	Easy	LeetCode	Solve on LeetCode
3	LR-130	Minimum Number of Days to Make m Bouquets	Hard	LeetCode	Solve on LeetCode

Pattern 21: Binary Search on Answer				3 Problems · Topic: Binary Search	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-127	Search in Rotated Sorted Array	Medium	LeetCode	Solve on LeetCode
2	LR-128	M bouquets	Medium	GeeksforGeeks	Solve on GFG
3	LR-129	kth in Multiplication Table	Medium	GeeksforGeeks	Solve on GFG

Pattern 22: Classic Binary Search				1 Problem · Topic: Binary Search	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-377	Binary Search	Easy	LeetCode	Solve on LeetCode

Pattern 23: Lower Bound / Upper Bound				1 Problem · Topic: Binary Search	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-378	Find First and Last Position of Element in Sorted Array	Medium	LeetCode	Solve on LeetCode

Pattern 24: Binary Search / Peak				1 Problem · Topic: Binary Search	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-379	Find Peak Element	Medium	LeetCode	Solve on LeetCode

Pattern 25: Linked List				14 Problems · Topic: Linked List	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-131	Linked List Cycle	Easy	LeetCode	Solve on LeetCode
2	LR-132	Reverse Linked List	Easy	LeetCode	Solve on LeetCode
3	LR-133	Linked List Cycle II	Medium	LeetCode	Solve on LeetCode
4	LR-134	Reverse Linked List II	Medium	LeetCode	Solve on LeetCode
5	LR-135	Reverse a Doubly	Medium	GeeksforGeeks	Solve on GFG
6	LR-136	Reverse in Groups (K-group reversal)	Medium	GeeksforGeeks	Solve on GFG
7	LR-137	Add Two Numbers	Medium	GeeksforGeeks	Solve on GFG
8	LR-138	Sort 0s, 1s, 2s	Medium	GeeksforGeeks	Solve on GFG
9	LR-139	Length of the Loop	Medium	GeeksforGeeks	Solve on GFG
10	LR-140	Intersection Point	Medium	GeeksforGeeks	Solve on GFG

#	ID	Problem Name	Difficulty	Platform	Practice Link
11	LR-141	Merge two sorted linked lists	Medium	GeeksforGeeks	Solve on GFG
12	LR-142	Palindrome Linked List	Medium	GeeksforGeeks	Solve on GFG
13	LR-144	Clone with Random Pointer	Hard	GeeksforGeeks	Solve on GFG
14	LR-145	Flattening	Hard	GeeksforGeeks	Solve on GFG

Pattern 26: Advanced Linked List

2 Problems · Topic: Linked List

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-143	Rotate List	Medium	LeetCode	Solve on LeetCode
2	LR-146	Reverse Nodes in k-Group	Hard	LeetCode	Solve on LeetCode

Pattern 27: Stack

8 Problems · Topic: Stack

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-147	Valid Parentheses	Easy	LeetCode	Solve on LeetCode
2	LR-148	Min Stack	Easy	LeetCode	Solve on LeetCode
3	LR-149	Implement Queue using Stacks	Easy	GeeksforGeeks	Solve on GFG
4	LR-150	Implement Stack using Linked List	Easy	GeeksforGeeks	Solve on GFG
5	LR-152	Decode String	Medium	GeeksforGeeks	Solve on GFG
6	LR-153	Longest Valid	Medium	GeeksforGeeks	Solve on GFG
7	LR-154	Score of Balanced	Medium	GeeksforGeeks	Solve on GFG
8	LR-155	Sort	Medium	GeeksforGeeks	Solve on GFG

Pattern 28: Monotonic Stack

4 Problems · Topic: Stack

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-151	Next Greater Element I	Easy	LeetCode	Solve on LeetCode
2	LR-156	Daily Temperatures	Medium	LeetCode	Solve on LeetCode
3	LR-157	Next Greater Element II	Medium	LeetCode	Solve on LeetCode
4	LR-158	Largest Rectangle in Histogram	Hard	LeetCode	Solve on LeetCode

Pattern 29: Monotonic Stack / Stock Span

1 Problem · Topic: Stack

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-376	Online Stock Span	Medium	LeetCode	Solve on LeetCode

Pattern 30: Infix → Postfix

1 Problem · Topic: Stack

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-380	Infix to Postfix Conversion	Medium	GeeksforGeeks	Solve on GFG

Pattern 31: Queue / Deque

3 Problems · Topic: Queue / Deque

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-159	Implement Stack using Queues	Easy	GeeksforGeeks	Solve on GFG
2	LR-160	First non-repeating in a stream	Medium	GeeksforGeeks	Solve on GFG
3	LR-161	Deque	Medium	GeeksforGeeks	Solve on GFG

Pattern 32: Monotonic Deque

1 Problem · Topic: Queue / Deque

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-162	Longest Subarray with Difference $\leq X$	Hard	GeeksforGeeks	Solve on GFG

Pattern 33: Queue / Time Window Design

1 Problem · Topic: Queue / Deque

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-381	Design Hit Counter	Medium	LeetCode	Solve on LeetCode

Pattern 34: Recursion

3 Problems · Topic: Recursion / Backtracking

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-163	Factorial of a Number	Easy	GeeksforGeeks	Solve on GFG
2	LR-164	Check if Array is Sorted	Easy	GeeksforGeeks	Solve on GFG
3	LR-165	Reverse a String	Easy	GeeksforGeeks	Solve on GFG

Pattern 35: Backtracking

8 Problems · Topic: Recursion / Backtracking

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-166	Generate Parentheses	Medium	LeetCode	Solve on LeetCode
2	LR-167	Rat in a Maze	Medium	GeeksforGeeks	Solve on GFG
3	LR-168	Permutations	Medium	LeetCode	Solve on LeetCode
4	LR-169	Unique Generalized Abbreviations	Medium	LeetCode	Solve on LeetCode
5	LR-170	Max Subsets	Medium	GeeksforGeeks	Solve on GFG
6	LR-171	Subsets With Duplicates	Medium	LeetCode	Solve on LeetCode
7	LR-173	Knight's Tour	Hard	GeeksforGeeks	Solve on GFG
8	LR-174	N Queen Problem	Hard	LeetCode	Solve on LeetCode

Pattern 36: Constraint Backtracking

1 Problem · Topic: Recursion / Backtracking

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-172	Word Search	Medium	GeeksforGeeks	Solve on GFG

Pattern 37: Backtracking / Combinations

1 Problem · Topic: Recursion / Backtracking

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-374	Combination Sum	Medium	LeetCode	Solve on LeetCode

Pattern 38: Backtracking / Partitioning

1 Problem · Topic: Recursion / Backtracking

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-375	Palindrome Partitioning	Medium	LeetCode	Solve on LeetCode

Pattern 39: Bit Manipulation

12 Problems · Topic: Bit Manipulation

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-175	Complement of Base 10 Number	Easy	LeetCode	Solve on LeetCode
2	LR-176	Max XOR of Two	Easy	GeeksforGeeks	Solve on GFG
3	LR-177	Pairs with XOR less than K	Easy	GeeksforGeeks	Solve on GFG
4	LR-178	Missing Number using XOR	Easy	LeetCode	Solve on LeetCode
5	LR-179	Power of Two	Easy	LeetCode	Solve on LeetCode
6	LR-180	Number of 1 Bits	Easy	LeetCode	Solve on LeetCode
7	LR-181	Counting Bits	Easy	LeetCode	Solve on LeetCode
8	LR-182	Single Number	Easy	LeetCode	Solve on LeetCode
9	LR-183	Single Number III	Medium	LeetCode	Solve on LeetCode
10	LR-184	Single Number II	Medium	LeetCode	Solve on LeetCode
11	LR-185	Bitwise AND of Numbers Range	Medium	LeetCode	Solve on LeetCode
12	LR-186	Sum of Two Integers	Medium	LeetCode	Solve on LeetCode

Pattern 40: Bit Masking				1 Problem · Topic: Bit Manipulation	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-187	Subsets	Medium	LeetCode	Solve on LeetCode

Pattern 41: Heap / Priority Queue				11 Problems · Topic: Heap / Priority Queue	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-188	Check for Heap	Easy	GeeksforGeeks	Solve on GFG
2	LR-189	Heap Sort	Easy	GeeksforGeeks	Solve on GFG
3	LR-190	Kth Largest Element in an Array	Medium	LeetCode	Solve on LeetCode
4	LR-191	K Max sum combinations	Medium	GeeksforGeeks	Solve on GFG
5	LR-192	BST to Min Heap	Medium	GeeksforGeeks	Solve on GFG
6	LR-193	Top K Frequent Elements	Medium	LeetCode	Solve on LeetCode
7	LR-194	Top K Frequent Words	Medium	LeetCode	Solve on LeetCode
8	LR-195	K Closest Points to Origin	Medium	LeetCode	Solve on LeetCode
9	LR-196	Reorganize String	Medium	LeetCode	Solve on LeetCode
10	LR-199	Sort by Frequency	Hard	GeeksforGeeks	Solve on GFG
11	LR-200	The Skyline Problem	Hard	GeeksforGeeks	Solve on GFG

Pattern 42: Two Heaps				4 Problems · Topic: Heap / Priority Queue	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-197	Next Interval	Medium	LeetCode	Solve on LeetCode
2	LR-201	Sliding Window Median	Hard	LeetCode	Solve on LeetCode
3	LR-202	IPO	Hard	LeetCode	Solve on LeetCode
4	LR-203	Find Median from Data Stream	Hard	LeetCode	Solve on LeetCode

Pattern 43: K-Way Merge				3 Problems · Topic: Heap / Priority Queue	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-198	Find K Pairs with Smallest Sums	Medium	LeetCode	Solve on LeetCode
2	LR-204	Smallest Range: Smallest Range Covering Elements from K Lists	Hard	LeetCode	Solve on LeetCode
3	LR-205	Merge k Sorted Lists	Hard	LeetCode	Solve on LeetCode

Pattern 44: Binary Tree Traversal				8 Problems · Topic: Binary Trees	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-206	Binary Tree Level Order Traversal	Easy	LeetCode	Solve on LeetCode
2	LR-207	Right View	Easy	GeeksforGeeks	Solve on GFG
3	LR-208	Morris Inorder	Easy	GeeksforGeeks	Solve on GFG
4	LR-209	Postorder	Easy	GeeksforGeeks	Solve on GFG
5	LR-217	Binary Tree Zigzag Level Order Traversal	Medium	LeetCode	Solve on LeetCode
6	LR-218	Boundary Traversal	Medium	GeeksforGeeks	Solve on GFG
7	LR-219	Bottom View	Medium	GeeksforGeeks	Solve on GFG
8	LR-220	Vertical Traversal	Medium	GeeksforGeeks	Solve on GFG

Pattern 45: Tree DFS				9 Problems · Topic: Binary Trees	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-210	Smallest string starting from Leaf	Easy	LeetCode	Solve on LeetCode
2	LR-211	Balanced Binary Tree	Easy	LeetCode	Solve on LeetCode

#	ID	Problem Name	Difficulty	Platform	Practice Link
3	LR-212	Binary Tree Paths	Easy	LeetCode	Solve on LeetCode
4	LR-213	Diameter of Binary Tree	Easy	LeetCode	Solve on LeetCode
5	LR-214	Maximum Depth of Binary Tree	Easy	LeetCode	Solve on LeetCode
6	LR-221	Count Paths for a Sum	Medium	LeetCode	Solve on LeetCode
7	LR-227	LCA	Hard	GeeksforGeeks	Solve on GFG
8	LR-228	Spiral Level Order	Hard	GeeksforGeeks	Solve on GFG
9	LR-229	Binary Tree Maximum Path Sum	Hard	LeetCode	Solve on LeetCode

Pattern 46: Tree BFS

4 Problems · Topic: Binary Trees

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-215	Level Averages in a Binary Tree	Easy	LeetCode	Solve on LeetCode
2	LR-216	Reverse Level Order Traversal	Easy	LeetCode	Solve on LeetCode
3	LR-222	All Nodes Distance K in Binary tree	Medium	LeetCode	Solve on LeetCode
4	LR-223	Maximum width of Binary Tree	Medium	LeetCode	Solve on LeetCode

Pattern 47: Tree Construction

2 Problems · Topic: Binary Trees

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-224	Construct BT from Postorder and Inorder	Medium	LeetCode	Solve on LeetCode
2	LR-225	Construct BT from Preorder and Inorder	Medium	LeetCode	Solve on LeetCode

Pattern 48: LCA / Ancestor Patterns

2 Problems · Topic: Binary Trees

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-226	LCA of Binary Tree	Medium	LeetCode	Solve on LeetCode
2	LR-230	Kth Ancestor of a Tree Node	Hard	LeetCode	Solve on LeetCode

Pattern 49: Tree Serialization

1 Problem · Topic: Binary Trees

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-373	Serialize and Deserialize Binary Tree	Hard	LeetCode	Solve on LeetCode

Pattern 50: BST

11 Problems · Topic: BST

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-231	Insert into a BST	Medium	LeetCode	Solve on LeetCode
2	LR-232	LCA of BST	Medium	LeetCode	Solve on LeetCode
3	LR-233	Validate Binary Search Tree	Medium	LeetCode	Solve on LeetCode
4	LR-234	Merge two BSTs	Medium	GeeksforGeeks	Solve on GFG
5	LR-235	Sorted Array to BST	Medium	GeeksforGeeks	Solve on GFG
6	LR-236	Recover BST	Medium	GeeksforGeeks	Solve on GFG
7	LR-237	Count of Structurally Unique Binary Search Trees	Medium	LeetCode	Solve on LeetCode
8	LR-238	Structurally Unique Binary Search Trees	Medium	LeetCode	Solve on LeetCode
9	LR-239	2 Sum in BST	Medium	GeeksforGeeks	Solve on GFG
10	LR-240	Kth Smallest Element	Medium	GeeksforGeeks	Solve on GFG
11	LR-241	Construct BST from Preorder	Medium	LeetCode	Solve on LeetCode

Pattern 51: Trie					3 Problems · Topic: Trie
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-242	Longest Word With All Prefixes	Medium	GeeksforGeeks	Solve on GFG
2	LR-243	Implement Trie (Prefix Tree)	Medium	LeetCode	Solve on LeetCode
3	LR-244	Implement Trie II (Prefix Tree)	Medium	LeetCode	Solve on LeetCode

Pattern 52: Advanced Trie					3 Problems · Topic: Trie
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-245	Count of distinct substrings	Medium	GeeksforGeeks	Solve on GFG
2	LR-246	Search Suggestions System	Medium	GeeksforGeeks	Solve on GFG
3	LR-247	Palindrome Pairs	Hard	GeeksforGeeks	Solve on GFG

Pattern 53: Intervals					4 Problems · Topic: Intervals
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-248	Meeting Rooms	Medium	GeeksforGeeks	Solve on GFG
2	LR-249	Insert Interval	Medium	LeetCode	Solve on LeetCode
3	LR-250	Merge Intervals	Medium	LeetCode	Solve on LeetCode
4	LR-251	Non-Overlapping Intervals	Medium	GeeksforGeeks	Solve on GFG

Pattern 54: Greedy					6 Problems · Topic: Greedy
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-252	Assign Cookies	Easy	GeeksforGeeks	Solve on GFG
2	LR-253	Fractional Knapsack	Easy	GeeksforGeeks	Solve on GFG
3	LR-254	Jump Game	Medium	GeeksforGeeks	Solve on GFG
4	LR-255	Job Sequencing	Medium	GeeksforGeeks	Solve on GFG
5	LR-258	Min time to finish all jobs with given constraints	Hard	GeeksforGeeks	Solve on GFG
6	LR-259	Minimize cash flow	Hard	GeeksforGeeks	Solve on GFG

Pattern 55: Greedy + Heap					3 Problems · Topic: Greedy
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-256	Task Scheduler	Medium	LeetCode	Solve on LeetCode
2	LR-257	Minimum Cost to Connect Sticks	Medium	LeetCode	Solve on LeetCode
3	LR-260	Maximum Performance of a Team	Hard	LeetCode	Solve on LeetCode

Pattern 56: Graph BFS / DFS					5 Problems · Topic: Graph Traversal
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-261	BFS of Graph	Easy	GeeksforGeeks	Solve on GFG
2	LR-262	DFS of Graph	Easy	GeeksforGeeks	Solve on GFG
3	LR-265	Min Knight steps for target	Medium	GeeksforGeeks	Solve on GFG
4	LR-266	Minimum Height Trees	Medium	LeetCode	Solve on LeetCode
5	LR-271	Shortest cycle in an undirected graph	Medium	GeeksforGeeks	Solve on GFG

Pattern 57: Grid BFS / DFS					2 Problems · Topic: Graph Traversal
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-263	Flood Fill Algorithm	Easy	LeetCode	Solve on LeetCode
2	LR-267	Number of islands	Medium	LeetCode	Solve on LeetCode

Pattern 58: Graph Cycle Detection					1 Problem · Topic: Graph Traversal
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-264	Detect cycle in an undirected graph	Easy	GeeksforGeeks	Solve on GFG

Pattern 59: Bipartite Graph					1 Problem · Topic: Graph Traversal
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-268	Bipartite Graph	Medium	LeetCode	Solve on LeetCode

Pattern 60: Shortest Path — BFS					3 Problems · Topic: Graph Traversal
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-269	Shortest Path in Binary Matrix	Medium	LeetCode	Solve on LeetCode
2	LR-270	Word Ladder	Hard	LeetCode	Solve on LeetCode
3	LR-278	Dijkstra's Algorithm: Cheapest Flights Within K Stops	Medium	LeetCode	Solve on LeetCode

Pattern 61: Topological Sort					4 Problems · Topic: Graph Algorithms
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-272	Course Schedule	Medium	LeetCode	Solve on LeetCode
2	LR-273	Sequence Reconstruction	Medium	LeetCode	Solve on LeetCode
3	LR-274	Topological Sort	Medium	GeeksforGeeks	Solve on GFG
4	LR-282	Alien Dictionary	Hard	LeetCode	Solve on LeetCode

Pattern 62: Dijkstra					2 Problems · Topic: Graph Algorithms
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-275	Bellman Ford: Network Delay time	Medium	LeetCode	Solve on LeetCode
2	LR-276	Shortest path with one curved edge in an undirected Graph	Medium	GeeksforGeeks	Solve on GFG

Pattern 63: Bellman-Ford					1 Problem · Topic: Graph Algorithms
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-277	Shortest Path with Negative Edges	Medium	GeeksforGeeks	Solve on GFG

Pattern 64: Floyd-Warshall					2 Problems · Topic: Graph Algorithms
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-279	Floyd-Warshall: Find the City with Smallest Number of Neighbours at a Threshold Distance	Medium	LeetCode	Solve on LeetCode
2	LR-280	Floyd Warshall	Medium	GeeksforGeeks	Solve on GFG

Pattern 65: Graph Algorithms					1 Problem · Topic: Graph Algorithms
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-281	Total Spanning Trees	Medium	GeeksforGeeks	Solve on GFG

Pattern 66: 0-1 BFS					2 Problems · Topic: Graph Algorithms
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-283	Minimum Cost to Make at Least One Valid Path in a Grid	Hard	LeetCode	Solve on LeetCode
2	LR-284	Minimum Obstacle Removal to Reach Corner	Hard	LeetCode	Solve on LeetCode

Pattern 67: Union Find / DSU

6 Problems · Topic: DSU / MST

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-285	Kruskal's Algorithm: Minimum Cost to connect all Points	Medium	LeetCode	Solve on LeetCode
2	LR-286	Union	Medium	GeeksforGeeks	Solve on GFG
3	LR-287	Detect Cycle using DSU	Medium	GeeksforGeeks	Solve on GFG
4	LR-288	Number of connected components	Medium	GeeksforGeeks	Solve on GFG
5	LR-289	Number of Operations to Make Network Connected	Medium	LeetCode	Solve on LeetCode
6	LR-290	Redundant Connection	Medium	LeetCode	Solve on LeetCode

Pattern 68: Minimum Spanning Tree

2 Problems · Topic: DSU / MST

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-291	Kruskal's Minimum Spanning Tree	Medium	GeeksforGeeks	Solve on GFG
2	LR-292	Prim's Minimum Spanning Tree	Medium	GeeksforGeeks	Solve on GFG

Pattern 69: Matrix

6 Problems · Topic: Matrix / Grid

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-293	Queries in a Matrix	Medium	GeeksforGeeks	Solve on GFG
2	LR-294	Game of Life	Medium	LeetCode	Solve on LeetCode
3	LR-295	Max Area of Island	Medium	LeetCode	Solve on LeetCode
4	LR-296	Number of Closed Islands	Medium	LeetCode	Solve on LeetCode
5	LR-297	Set Matrix Zeroes	Medium	LeetCode	Solve on LeetCode
6	LR-298	Surrounded Regions	Medium	LeetCode	Solve on LeetCode

Pattern 70: Grid Algorithms

2 Problems · Topic: Matrix / Grid

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-299	Rotting Oranges	Medium	LeetCode	Solve on LeetCode
2	LR-300	Longest Increasing Path in a Matrix	Hard	LeetCode	Solve on LeetCode

Pattern 71: 1D DP

5 Problems · Topic: Dynamic Programming

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-301	Fibonacci Number	Easy	LeetCode	Solve on LeetCode
2	LR-304	Derangements	Medium	GeeksforGeeks	Solve on GFG
3	LR-305	Possible Decodings	Medium	GeeksforGeeks	Solve on GFG
4	LR-306	Coin Change	Medium	LeetCode	Solve on LeetCode
5	LR-307	Interleaving String	Medium	GeeksforGeeks	Solve on GFG

Pattern 72: DP Fundamentals

2 Problems · Topic: Dynamic Programming

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-302	Longest Common Subsequence	Medium	LeetCode	Solve on LeetCode
2	LR-303	Sorted subsequence of 3	Medium	GeeksforGeeks	Solve on GFG

Pattern 73: Knapsack DP				5 Problems · Topic: Dynamic Programming	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-308	House Robber ii	Medium	LeetCode	Solve on LeetCode
2	LR-309	Target Sum	Medium	LeetCode	Solve on LeetCode
3	LR-310	Coin Change II	Medium	LeetCode	Solve on LeetCode
4	LR-311	Minimum Cost For Tickets	Medium	LeetCode	Solve on LeetCode
5	LR-312	Perfect Squares	Medium	LeetCode	Solve on LeetCode

Pattern 74: Subsequence DP				6 Problems · Topic: Dynamic Programming	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-313	0/1 Knapsack	Medium	GeeksforGeeks	Solve on GFG
2	LR-314	Maximum Length of Pair Chain	Medium	LeetCode	Solve on LeetCode
3	LR-315	Number of LIS	Medium	LeetCode	Solve on LeetCode
4	LR-316	Rod Cutting	Medium	GeeksforGeeks	Solve on GFG
5	LR-317	Consecutive Subsequence	Medium	GeeksforGeeks	Solve on GFG
6	LR-318	Longest Consecutive Subsequence	Medium	GeeksforGeeks	Solve on GFG

Pattern 75: String DP				3 Problems · Topic: Dynamic Programming	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-319	Longest Common Substring	Medium	GeeksforGeeks	Solve on GFG
2	LR-320	Check Subsequence	Medium	GeeksforGeeks	Solve on GFG
3	LR-321	Longest Palindromic Subsequence	Medium	LeetCode	Solve on LeetCode

Pattern 76: Grid DP				6 Problems · Topic: Dynamic Programming	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-322	Largest Square Submatrix with All 1s	Medium	GeeksforGeeks	Solve on GFG
2	LR-323	Longest Increasing Path	Medium	GeeksforGeeks	Solve on GFG
3	LR-324	Maximum path sum in matrix	Medium	GeeksforGeeks	Solve on GFG
4	LR-325	Dungeon Game: Dungeon Game	Medium	LeetCode	Solve on LeetCode
5	LR-326	Maximal Square	Medium	LeetCode	Solve on LeetCode
6	LR-327	Minimum cost Path	Medium	GeeksforGeeks	Solve on GFG

Pattern 77: Stock DP				4 Problems · Topic: Dynamic Programming	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-328	Buy and Sell Stocks iii	Medium	LeetCode	Solve on LeetCode
2	LR-329	Buy and Sell Stocks iv	Medium	LeetCode	Solve on LeetCode
3	LR-330	Buy and Sell - k Transactions	Medium	GeeksforGeeks	Solve on GFG
4	LR-331	Buy and Sell with Transaction Fee	Medium	GeeksforGeeks	Solve on GFG

Pattern 78: Partition DP				5 Problems · Topic: Dynamic Programming	
#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-332	Matrix Chain Multiplication	Medium	GeeksforGeeks	Solve on GFG
2	LR-333	Boolean Parenthesization	Medium	GeeksforGeeks	Solve on GFG
3	LR-334	Optimal Strategy	Medium	GeeksforGeeks	Solve on GFG
4	LR-335	Minimum Cost to Cut a Stick	Medium	LeetCode	Solve on LeetCode
5	LR-340	Burst Balloons	Hard	LeetCode	Solve on LeetCode

Pattern 79: Tree DP

4 Problems · Topic: Dynamic Programming

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-336	House Robber III	Medium	LeetCode	Solve on LeetCode
2	LR-337	Distribute Coins in Binary Tree	Medium	LeetCode	Solve on LeetCode
3	LR-338	Longest ZigZag Path in a Binary Tree	Medium	LeetCode	Solve on LeetCode
4	LR-341	Binary Tree Cameras	Hard	LeetCode	Solve on LeetCode

Pattern 80: Bitmask DP

4 Problems · Topic: Dynamic Programming

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-339	Partition to K Equal Sum Subsets	Medium	LeetCode	Solve on LeetCode
2	LR-342	Smallest Sufficient Team	Hard	LeetCode	Solve on LeetCode
3	LR-343	Shortest Path Visiting All Nodes	Hard	LeetCode	Solve on LeetCode
4	LR-344	Traveling Salesman Problem	Hard	GeeksforGeeks	Solve on GFG

Pattern 81: Segment Tree / Fenwick Tree

5 Problems · Topic: Segment Tree / Fenwick Tree

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-345	Range Sum Query - Mutable	Medium	LeetCode	Solve on LeetCode
2	LR-346	Range Sum Query 2D - Mutable	Hard	LeetCode	Solve on LeetCode
3	LR-347	Count of Smaller Numbers After Self	Hard	LeetCode	Solve on LeetCode
4	LR-348	Longest Increasing Subsequence II	Hard	LeetCode	Solve on LeetCode
5	LR-349	My Calendar III	Hard	LeetCode	Solve on LeetCode

Pattern 82: Design

8 Problems · Topic: Design

#	ID	Problem Name	Difficulty	Platform	Practice Link
1	LR-350	Design HashMap	Easy	LeetCode	Solve on LeetCode
2	LR-351	LRU Cache	Medium	LeetCode	Solve on LeetCode
3	LR-352	Design Circular Queue	Medium	LeetCode	Solve on LeetCode
4	LR-353	Design Twitter	Medium	LeetCode	Solve on LeetCode
5	LR-354	Insert Delete GetRandom O(1)	Medium	LeetCode	Solve on LeetCode
6	LR-355	Design Add and Search Words Data Structure	Medium	LeetCode	Solve on LeetCode
7	LR-356	Time Based Key-Value Store	Medium	LeetCode	Solve on LeetCode
8	LR-357	LFU Cache	Hard	LeetCode	Solve on LeetCode